

MAT 5173 — Problem Set 2

Due Wednesday 30 September 2026, at the start of the Q&A session.

This set covers the three meetings of 21, 23 and 28 September, that is, Ash Ch. 2 from its basic definitions through unique factorization: rings, subrings, units and zero divisors (§2.1), ideals, homomorphisms and quotient rings (§2.2), the isomorphism theorems for rings (§2.3), maximal and prime ideals (§2.4), polynomial rings (§2.5), and unique factorization (§2.6). The vocabulary of §2.1 is used throughout rather than examined on its own; the problems below take up §2.2 onward. (Li-Zhao §4.1, §4.5, §4.7 and §5.1 carry the same material at a gentler pace, should you want a second exposition.) Note that nothing on this set reaches past unique factorization: principal ideal domains, Euclidean domains and rings of fractions (§2.7–§2.8) come *after* Midterm 1, and are therefore not examinable on it.

Maximum six pages. Each problem is scored on *completeness* alone, not on correctness: **2** for a nominally complete solution, **1** for a clearly partial one, **0** for nothing submitted or for filler with no substance. Any submission containing non-human-readable content, stray characters, or unproofread AI output scores 0 for that item.

Sources. AI assistance, solution keys (Ash prints them at the back), collaboration and the internet are all permitted, without restriction. Include one line naming what you used. Note that at the Q&A session we name the problem you are to explain; you do not choose, so be ready on every problem on this set.

Notation. We follow Ash throughout. Ash assumes commutativity only where he needs it—in §2.4 and §2.6, not in §2.2–§2.3—and we do likewise: each problem below states its own standing hypotheses, and you should not import commutativity into a problem that does not grant it. All rings have an identity, written 1; an *ideal* means a two-sided ideal unless the ring at hand is commutative, in which case the distinction is vacuous. We write $(a_1, \dots, a_k)$ for the ideal generated by $a_1, \dots, a_k$, and R/I for the quotient ring. An element u of a ring is a *unit* if it has a multiplicative inverse there; note that this depends on the ring, not on the element alone (a point Problem 4 turns on).

Problem 1 (ideals, quotient rings and the isomorphism theorems) — §2.2–§2.3

Rings here are **not** assumed commutative, save where a specific example is.

1. Let $\varphi: R \rightarrow S$ be a ring homomorphism. Prove that $\ker \varphi$ is an ideal of R , that $\operatorname{im} \varphi$ is a subring of S , and that φ is injective if and only if $\ker \varphi = \{0\}$. (Observe where two-sidedness of $\ker \varphi$ is used, and note that $\operatorname{im} \varphi$ is in general *not* an ideal of S ; give an example witnessing this.)
2. Prove that evaluation at $\sqrt{-1}$, namely

$$\varepsilon: \mathbb{R}[x] \rightarrow \mathbb{C}, \quad \varepsilon(f) := f(\sqrt{-1}),$$

is a surjective ring homomorphism with kernel exactly $(x^2 + 1)$, and deduce that

$$\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}.$$

(For the kernel you may divide by the monic polynomial $x^2 + 1$, as in §2.5; name the isomorphism theorem you invoke.)

3. Let m and n be positive integers with $n \geq 2$ and $m \mid n$. Prove that $m\mathbb{Z}/n\mathbb{Z}$ is an ideal of $\mathbb{Z}/n\mathbb{Z}$ and that

$$(\mathbb{Z}/n\mathbb{Z})/(m\mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/m\mathbb{Z}$$

as rings. You proved the corresponding statement for groups in Problem 4(2) of Set 1; say precisely what must be added to that argument, and what may be quoted from it verbatim.

Problem 2 (maximal and prime ideals) — §2.4

Throughout this problem R is commutative with identity, as in Ash §2.4, and $I \subsetneq R$ is a proper ideal.

1. Prove that I is prime if and only if R/I is an integral domain, and that I is maximal if and only if R/I is a field. Deduce that every maximal ideal is prime. (The deduction should be one sentence; that is the point of proving the two characterizations together.)
 2. Work both parts of the converse in $\mathbb{Z}[x]$. Show that (x) is prime but not maximal, and that $(2, x)$ is maximal. In each case identify the quotient ring explicitly—that is the whole labour, given part 1—.
 3. Prove that a finite integral domain is a field. Conclude that in a finite commutative ring with identity every prime ideal is maximal; hence the phenomenon exhibited in part 2 is unavailable in the finite world.
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Problem 3 (polynomial rings) — §2.5

Let R be a commutative ring with identity.

1. Let $g \in R[x]$ be monic of degree $m \geq 1$. Prove that for every $f \in R[x]$ there exist *unique* $q, r \in R[x]$ with

$$f = qg + r, \quad \deg r < m$$

(with the convention $\deg 0 := -\infty$). Note that no hypothesis beyond monicity of g is needed; in particular R need not be a field, nor even a domain.

2. Deduce the factor theorem: for $a \in R$, we have $f(a) = 0$ if and only if $(x - a) \mid f$. Deduce further that if R is an integral domain, then a nonzero $f \in R[x]$ of degree n has at most n roots in R .
 3. Exhibit a polynomial of degree 2 over $\mathbb{Z}/8\mathbb{Z}$ having four roots. Identify exactly which step of your proof of part 2 fails there, and why the failure is one of hypothesis rather than of argument.
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Problem 4 (unique factorization, and how it fails) — §2.6

Let

$$R := \mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} : a, b \in \mathbb{Z}\},$$

a subring of $\mathbb{C}$, hence an integral domain, and define the *norm*

$$N(a + b\sqrt{-5}) := a^2 + 5b^2 \in \mathbb{Z}_{\geq 0}.$$

Recall that α is *irreducible* if it is a nonzero nonunit whose only factorizations $\alpha = \beta\gamma$ have β or γ a unit, and *prime* if it is a nonzero nonunit with $\alpha \mid \beta\gamma$ implying $\alpha \mid \beta$ or $\alpha \mid \gamma$.

1. Prove that N is multiplicative, i.e. $N(\alpha\beta) = N(\alpha)N(\beta)$ for all $\alpha, \beta \in R$. Deduce that α is a unit of R if and only if $N(\alpha) = 1$, and hence that the units of R are exactly ± 1 .
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2. Prove that 2, 3, $1 + \sqrt{-5}$ and $1 - \sqrt{-5}$ are all irreducible in R . (The norms to rule out are small; note that $N(\alpha) = 2$ and $N(\alpha) = 3$ are both impossible.)
3. Conclude that R is not a unique factorization domain, by exhibiting two factorizations of 6 into irreducibles that are not rearrangements of one another up to units. Show moreover that 2 is irreducible but *not* prime in R ; hence the implication “irreducible $\Rightarrow$ prime”, which holds in any unique factorization domain, genuinely fails here. (This is the example to keep in mind whenever you are tempted to treat the two words as synonyms.)